

Teo aut disco unidad composicion

Teorema 1. Sean $f = \rho_{\gamma_1} \circ \tau_{a_1}$ y $g = \rho_{\gamma_2} \circ \tau_{a_2}$ en $\text{Aut}(\mathbb{D})$, entonces

$$g \circ f = \rho_{\gamma} \circ \tau_a \quad \text{donde} \quad a = \tau_{a_1}(a_2) = \frac{a_1 - a_2}{1 - \overline{a_1}a_2} \quad \wedge \quad \gamma = \begin{cases} \gamma_1 \tau_{\overline{a_1}a_2}(\gamma_2) & \text{si } a_2 \neq a_1\gamma_1, \\ -\gamma_1\gamma_2 & \text{si } a_2 = a_1\gamma_1, \end{cases}$$

En particular, si $a_2 = a_1\gamma_1$, entonces $g \circ f = \rho_{\gamma_1\gamma_2}$.

Demostración: Operando: $g \circ f = \rho_{\gamma_2} \circ (\tau_{a_2} \circ \rho_{\gamma_1}) \circ \tau_{a_1} \stackrel{\text{lem-involucion-disco-unidad-rho/1}}{=} \rho_{\gamma_2} \circ \rho_{\gamma_1} \circ (\tau_{\overline{\gamma_1}a_2} \circ \tau_{a_1})$.

Ahora bien, por el `lem-aut-disco-unidad-composicion-involuciones/lema 1`, tenemos que $\tau_{\overline{\gamma_1}a_2} \circ \tau_{a_1} = \rho_{\tilde{\gamma}} \circ \tau_{\tilde{a}}$ donde

$$\tilde{\gamma} = \begin{cases} \tau_{\overline{a_1}a_2\overline{\gamma_2}}(1) & \text{si } a_1 \neq \overline{\gamma_1}a_2, \\ -1 & \text{si } a_1 = \overline{\gamma_1}a_2, \end{cases} \quad \wedge \quad \tilde{a} = \tau_{a_1}(\overline{\gamma_1}a_2) = \frac{a_1 - \overline{\gamma_1}a_2}{1 - \overline{a_1}\gamma_1a_2}.$$

Concluimos por tanto que si $g \circ f = \rho_{\gamma} \circ \tau_a$, entonces

$$\gamma = \gamma_1\gamma_2\tilde{\gamma} = \begin{cases} \gamma_1\gamma_2 \cdot \tau_{\overline{a_1}a_2\overline{\gamma_2}}(1) & \text{si } a_2 \neq a_1\gamma_1, \\ -\gamma_1\gamma_2 & \text{si } a_2 = a_1\gamma_1, \end{cases} \quad \wedge \quad a = \tilde{a} = \tau_{a_1}(\overline{\gamma_1}a_2).$$

Finalmente, tenemos que $\gamma_2 \cdot \tau_{\overline{a_1}a_2\overline{\gamma_2}}(1) = \gamma_2 \cdot \frac{\overline{a_1}a_2\overline{\gamma_2} - 1}{1 - a_1\overline{a_2}\gamma_2} = \frac{\overline{a_1}a_2 - \gamma_2}{1 - a_1\overline{a_2}\gamma_2} = \tau_{\overline{a_1}a_2}(\gamma_2)$ y se sigue el resultado deseado. ■